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The Frozen River

DOES TIME FLOW?

Time is among the most familiar yet least understood concepts that humanity has ever encountered. We say that it flies, we say that it's money, we try to save it, we get annoyed when we waste it. But what is time? To paraphrase St. Augustine and Justice Potter Stewart, we know it when we see it, but surely, at the dawn of the third millennium our understanding of time must be deeper than that. In some ways, it is. In other ways, it's not. Through centuries of puzzling and pondering, we have gained insight into some of time's mysteries, but many remain. Where does time come from? What would it mean to have a universe without time? Could there be more than one time dimension, just as there is more than one space dimension? Can we "travel" to the past? If we did, could we change the subsequent unfolding of events? Is there an absolute, smallest amount of time? Is time a truly fundamental ingredient in the makeup of the cosmos, or simply a useful construct to organize our perceptions, but one not found in the lexicon with which the most fundamental laws of the universe are written? Could time be a derivative notion, emerging from some more basic concept that has yet to be discovered?

Finding complete and fully convincing answers to these questions ranks among the most ambitious goals of modern science. Yet the big questions are by no means the only ones. Even the everyday experience of time taps into some of the universe's thorniest conundrums.

Time and Experience

Special and general relativity shattered the universality, the oneness, of time. These theories showed that we each pick up a shard of Newton's